

NATALIA BURGIER

HM # 3

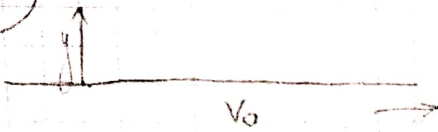
2,3 on grade

28/30

LTC SL

Extra credit : 1/2

1



$t=0$

Postulates

$$1) \begin{aligned} V_x &= V_x(y, t) \\ V_y &= 0 \\ V_z &= 0 \end{aligned}$$

2) constant ρ and μ (properties)

3) ignore gravity

Continuity eq 3.4-1 $\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho V_x) + \frac{\partial}{\partial y}(\rho V_y) + \frac{\partial}{\partial z}(\rho V_z) = 0$
 $0 = 0$

3.6-1 x:

$$\rho \left(\frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} + V_y \frac{\partial V_x}{\partial y} + V_z \frac{\partial V_x}{\partial z} \right) = - \frac{\partial P}{\partial x} + \mu \left[\frac{\partial^2 V_x}{\partial x^2} + \frac{\partial^2 V_x}{\partial y^2} + \frac{\partial^2 V_x}{\partial z^2} \right] + \rho g_x$$

$$\rho \frac{\partial V_x}{\partial t} = - \frac{\partial P}{\partial x} + \mu \frac{\partial^2 V_x}{\partial y^2} \quad \dots (1)$$

3.6-2 y:

$$\rho \left(\frac{\partial V_y}{\partial t} + V_x \frac{\partial V_y}{\partial x} + V_y \frac{\partial V_y}{\partial y} + V_z \frac{\partial V_y}{\partial z} \right) = - \frac{\partial P}{\partial y} + \mu \left[\frac{\partial^2 V_y}{\partial x^2} + \frac{\partial^2 V_y}{\partial y^2} + \frac{\partial^2 V_y}{\partial z^2} \right] + \rho g_y$$

$$0 = - \frac{\partial P}{\partial y} \quad \dots (2)$$

3.6-3 z:

$$\rho \left(\frac{\partial V_z}{\partial t} + V_x \frac{\partial V_z}{\partial x} + V_y \frac{\partial V_z}{\partial y} + V_z \frac{\partial V_z}{\partial z} \right) = - \frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 V_z}{\partial x^2} + \frac{\partial^2 V_z}{\partial y^2} + \frac{\partial^2 V_z}{\partial z^2} \right] + \rho g_z$$

$$0 = - \frac{\partial P}{\partial z} \quad \dots (3)$$

From (1) (2) (3) $\Rightarrow P = P(x)$ P constant w.r.t y and z
 as $y \rightarrow \infty$ $P = P_0$

$\frac{\partial P}{\partial x} = 0 \Rightarrow$ pressure is constant in all directions

Therefore the only left eq is:

$$\frac{\partial V_x}{\partial t} = \frac{\mu}{\rho} \frac{\partial^2 V_x}{\partial y^2}$$

$$\frac{\partial V_x}{\partial t} = \nu \frac{\partial^2 V_x}{\partial y^2}$$

$$\frac{\partial^2 \phi}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial \eta} \frac{1}{\sqrt{4\nu t}} \right) = \frac{1}{\sqrt{4\nu t}} \frac{\partial}{\partial y} \frac{\partial \phi}{\partial \eta} = \frac{\partial^2 \phi}{\partial \eta^2} \left(\frac{1}{\sqrt{4\nu t}} \right)$$

$$= \frac{\partial^2 \phi}{\partial \eta^2} \left(\frac{1}{4\nu t} \right)$$

Plugging into gives:

$$-\frac{1}{2} \frac{\partial \phi}{\partial \eta} = \sqrt{\left(\frac{\partial^2 \phi}{\partial \eta^2} \frac{1}{4\nu t} \right)}$$

$$0 = \sqrt{\left(\frac{\partial^2 \phi}{\partial \eta^2} \frac{1}{4\nu t} \right)} + \frac{1}{2} \frac{\partial \phi}{\partial \eta}$$

$$0 = \frac{\partial^2 \phi}{\partial \eta^2} + 4 \frac{1}{2} \frac{\partial \phi}{\partial \eta}$$

$$0 = \frac{\partial^2 \phi}{\partial \eta^2} + 2 \eta \frac{\partial \phi}{\partial \eta}$$

New

B.C #1 $\phi = 1$ @ $\eta = 0$ from B.C #1

B.C #2 $\phi = 0$ @ $\eta = \infty$ from B.C #2 and I.C

App C-1.8

$$\frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0 \xrightarrow{\text{Sol.}} y = C_1 \int_0^x \exp(-\bar{x}^2) d\bar{x} + C_2$$

$$\phi = C_1 \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta} + C_2$$

Apply B.C #1

$$1 = C_1 \int_0^0 \exp(0) d\bar{\eta} + C_2 \Rightarrow C_2 = 1$$

Apply B.C #2

$$0 = C_1 \int_0^\infty e^{-\bar{\eta}^2} d\bar{\eta} + 1$$

$$0 = C_1 \frac{\sqrt{\pi}}{2} + 1$$

$$-1 = C_1 \frac{\sqrt{\pi}}{2}$$

$$-\frac{2}{\sqrt{\pi}} = C_1$$

$$\phi = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta}$$

$$2) \mu = \int \frac{\partial^2 V}{\partial y^2} dy = \int -\rho g dy$$

$$\int \frac{\partial V}{\partial y} dy = \int \frac{-\rho g y}{\mu} dy + C_1$$

$$V = -\frac{\rho g y^2}{2\mu} + C_1 y + C_2$$

B.C #1 $V=0$ @ $y=0$

B.C #2 $\frac{dv}{dy} = 0$ @ $y=\delta$

Apply B.C #1

$$0 = 0 + 0 = C_2$$

Apply B.C #2

$$0 = -\frac{\rho g \delta}{\mu} + C_1$$

$$\frac{\rho g \delta}{\mu} = C_1$$

$$V = -\frac{\rho g y^2}{2\mu} + \frac{y \rho g \delta}{\mu} = \frac{\rho g \delta^2}{2\mu} \left[2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right]$$

B.C #3 $V_2 = V_{2max}$ @ $y=\delta$

$$V_{2max} = \frac{\rho g \delta^2}{2\mu}$$

$$V_2 = V_{2max} \left[1 - \left(\frac{x}{\delta}\right)^2 \right] = V_{2max} \left[1 - \left(1 - \frac{y}{\delta}\right)^2 \right] = V_{2max} \left[1 - 1 + 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right]$$

$$V_2 = 2V_{2max} \left(\frac{y}{\delta}\right) = 2 \frac{\rho g \delta^2}{2\mu} \left(\frac{y}{\delta}\right) = \frac{\rho g \delta}{\mu} y$$

b) Assumptions: - laminar

$$- v_y = 0$$

$$v_x = 0$$

$$v_z = V$$

$$\frac{\partial T}{\partial x} = 0$$

$$\frac{\partial^2 T}{\partial z^2} = 0$$

$$\rho \hat{C}_p v_z \frac{dT}{dz} = k \frac{\partial^2 T}{\partial y^2}$$

$$v_z = \frac{\rho g \delta y}{\mu}$$

$$\rho \hat{C}_p \frac{\rho g \delta y}{\mu} \frac{dT}{dz} = k \frac{\partial^2 T}{\partial y^2}$$

$$\beta = \frac{\mu k}{\rho^2 \hat{C}_p g \delta} \Rightarrow \frac{dT}{dz} y = \beta \frac{\partial^2 T}{\partial y^2}$$

$$\frac{k \mu}{\rho^2 g \delta \hat{C}_p}$$

c) INFINITE SLAB condition can be used

- $T = T_1$ @ $y = 0$ for $z > 0$ (

- $T = T_0$ @ $y \rightarrow \infty$ for z is finite (

- $T = T_0$ @ $y > 0$ for $z = 0$ (not contact with the plate yet

$$\theta = 0 \quad \eta \rightarrow \infty$$

$$\theta = 1 \quad \eta = 0$$

$$\frac{d^2 \theta}{d\eta^2} + 3\eta^2 \frac{d\theta}{d\eta} = 0$$

$$y \frac{\partial \theta}{\partial z} = \beta \frac{\partial^2 \theta}{\partial y^2} \quad (12.3.4-3)$$

$$\frac{\partial \theta}{\partial z} = \frac{\partial \theta}{\partial \eta} \frac{\partial \eta}{\partial z} = \frac{d\theta}{d\eta} \frac{y}{\sqrt[3]{9\beta z}} \left(-\frac{1}{3z}\right) = -\eta \frac{1}{32}$$

$$\frac{\partial \theta}{\partial y} = \frac{\partial \theta}{\partial \eta} \frac{\partial \eta}{\partial y} = \frac{d\theta}{d\eta} \frac{1}{\sqrt[3]{9\beta z}}$$

$$\frac{\partial^2 \theta}{\partial y^2} = \frac{d}{d\eta} \left(\frac{d\theta}{d\eta} \frac{1}{\sqrt[3]{9\beta z}} \right) \frac{\partial \eta}{\partial y} = \frac{d^2 \theta}{d\eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right)^2 = \frac{d^2 \theta}{d\eta^2} \left(\frac{\eta^2}{y^2} \right)$$

$$y \left(\frac{d\theta}{d\eta} \frac{-1}{32} \frac{y}{\sqrt[3]{9\beta z}} \right) = \beta \left(\frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{\eta^2}{y^2} \right) \right)$$

$$0 = \beta \left(\frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right)^2 \right) + \frac{1}{32} \frac{y^2}{\sqrt[3]{9\beta z}} \frac{d\theta}{d\eta}$$

$$0 = \beta \frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right)^2 + \frac{1}{32} \frac{y^2}{\sqrt[3]{9\beta z}} \frac{d\theta}{d\eta}$$

$$0 = \beta \frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right) + \frac{1}{32} y^2 \frac{d\theta}{d\eta}$$

$$0 = \beta \frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{\eta^2}{y^2} \right) + \frac{1}{32} y \eta \frac{d\theta}{d\eta}$$

$$\theta = \int_0^\eta \exp(-\eta^{-3}) d\eta$$

$$0 = \frac{d^2 \theta}{d\eta^2} + 3\eta^2 \frac{d\theta}{d\eta}$$

$$2) \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2} \quad / \cdot \left(-k \frac{\partial}{\partial y}\right)$$

$$\frac{\partial T}{\partial t} \left(-k \frac{\partial}{\partial y}\right) = \alpha \frac{\partial^2 T}{\partial y^2} \left(-k \frac{\partial}{\partial y}\right)$$

$$\frac{\partial}{\partial t} \left(-k \frac{\partial T}{\partial y}\right) = \alpha \frac{\partial^2}{\partial y^2} \left(-k \frac{\partial T}{\partial y}\right)$$

$$\text{where } -k \frac{\partial T}{\partial y} = q_y$$

$$\frac{\partial q_y}{\partial t} = \alpha \frac{\partial^2 q_y}{\partial y^2}$$

$$\text{I.C. \#1 } q_y(y, 0) = 0$$

$$\text{B.C. \#1 } q_y(0, t) = q_0 \text{ (constant)}$$

$$\text{B.C. \#2 } q_y(\infty, t) = 0$$

dimensionless variables

$$\theta = \frac{q_y}{q_0} \quad \eta = \frac{y}{\sqrt{4\alpha t}} \quad \text{assuming } \theta = \theta(\eta)$$

$$\text{I.C. \#1 } \theta(0) = 0$$

$$\text{B.C. \#2 } \theta(\infty, t) = 0$$

$$\text{B.C. \#1 } \theta(0, t) = 1$$

$$\theta(y, t; \alpha)$$

$$\frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial y^2}$$

$$* \quad \frac{\partial \theta}{\partial t} = \frac{\partial \theta}{\partial \eta} \frac{\partial \eta}{\partial t} \rightarrow \text{Arise}$$

$$\begin{aligned} \frac{\partial \eta}{\partial t} &= \frac{\partial}{\partial t} \left(\frac{y}{\sqrt{4\alpha t}} \right) = \frac{y}{\sqrt{4\alpha}} \frac{\partial}{\partial t} t^{-\frac{1}{2}} = \\ &= \frac{y}{\sqrt{4\alpha}} \left(-\frac{1}{2} t^{-\frac{3}{2}} \right) = -\frac{1}{2} \frac{y}{\sqrt{4\alpha}} t^{-1} t^{-\frac{1}{2}} = \\ &= -\frac{1}{2} \frac{y}{(\sqrt{4\alpha t}) t} = -\frac{\eta}{2t} \end{aligned}$$

$$\frac{\partial \theta}{\partial t} = -\frac{\eta}{2t} \frac{\partial \theta}{\partial \eta}$$

$$* \quad \frac{\partial \theta}{\partial y} = \frac{\partial \theta}{\partial \eta} \frac{\partial \eta}{\partial y} \rightarrow \text{Arise}$$

$$\frac{\partial \eta}{\partial y} = \frac{\partial}{\partial y} \left(\frac{y}{\sqrt{4\alpha t}} \right) = \frac{1}{\sqrt{4\alpha t}} \frac{\partial}{\partial y} y = \frac{1}{\sqrt{4\alpha t}}$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{\sqrt{4\alpha t}} \frac{\partial \theta}{\partial \eta}$$

$$= \frac{\partial}{\partial y} \left(\frac{\partial \theta}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{1}{\sqrt{4\alpha t}} \frac{\partial \theta}{\partial \eta} \right) = \frac{1}{\sqrt{4\alpha t}} \frac{\partial}{\partial y} \frac{\partial \theta}{\partial \eta} = \frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{1}{\sqrt{4\alpha t}} \right)^2 =$$

$$\frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{1}{4\alpha t} \right)$$

Plugging into $\frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial y^2}$

$$-\frac{\eta}{2t} \frac{\partial \theta}{\partial \eta} = \alpha \frac{1}{4\alpha t} \frac{\partial^2 \theta}{\partial \eta^2}$$

$$0 = \left(\alpha \frac{1}{4\alpha t} \frac{\partial^2 \theta}{\partial \eta^2} \right) + \frac{\eta}{2t} \frac{\partial \theta}{\partial \eta} \quad / -4t$$

$$0 = \frac{\partial^2 \theta}{\partial \eta^2} + 2\eta \frac{\partial \theta}{\partial \eta}$$

New

B.C #1 $\theta = 1$

B.C #2 $\theta = 0$

① $\eta = 0$

② $\eta = \infty$

for B.C #1

for B.C #2 and 1C

App. C-18

$$\frac{\partial^2 y}{\partial x^2} + 2x \frac{\partial y}{\partial x} = 0 \quad \text{Sol.} \quad y = C_1 \int_0^x \exp(-\bar{x}^2) d\bar{x} + C_2$$

$$\theta = C_1 \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta} + C_2$$

Apply B.C #1

$$1 = C_1 \int_0^0 \exp(-\bar{\eta}^2) d\bar{\eta} + C_2 \Rightarrow C_2 = 1$$

Apply B.C #2

$$0 = C_1 \underbrace{\int_0^\infty \exp(-\bar{\eta}^2) d\bar{\eta}}_{\frac{\sqrt{\pi}}{2}} + 1$$

$$-1 = C_1 \frac{\sqrt{\pi}}{2}$$

$$-\frac{2}{\sqrt{\pi}} = C_1$$

$$\theta = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta} = 1 - \text{erf}(\eta) = \text{erfc}(\eta)$$

$$\therefore \frac{\partial y}{\partial x} = \frac{2}{\sqrt{\pi}} \int_\eta^\infty (e^{-\bar{\eta}^2}) d\bar{\eta}$$

$$\psi_0 = \psi_0 \left(\frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-\eta^2} d\eta \right)$$

using Fourier's law

$$\psi_0 = -k \frac{\partial T}{\partial y}$$

$$-k \frac{\partial T}{\partial y} = \psi_0 (\operatorname{erfc}(\eta)) / -k$$

$$\frac{\partial T}{\partial y} = \frac{\psi_0}{k} \operatorname{erfc}(\eta)$$

$$\int_T^{\infty} \partial T = -\frac{\psi_0}{k} \int_0^{\infty} \operatorname{erfc}(\eta) d\eta \quad -k \int_T^{\infty} \partial T = \int_0^{\infty} \psi_0 (\eta) d\eta$$

$$T_0 - T = -\frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \int_0^{\infty} \int_0^{\infty} e^{-\eta^2} d\eta d\bar{y} =$$

$$= \frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \int_0^{\infty} \int_0^{\infty} e^{-\eta^2} d\eta d\bar{y} =$$

$$= \frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \sqrt{h_{\text{ext}}} \int_0^{\infty} \int_0^{\infty} e^{-\eta^2} d\eta d\psi =$$

$$= \frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \sqrt{h_{\text{ext}}} \int_0^{\infty} \int_0^{\infty} e^{-\eta^2} d\psi d\eta =$$

$$= \frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \sqrt{h_{\text{ext}}} \int_0^{\infty} e^{-\eta^2} \left[\eta - \frac{\eta}{\sqrt{\pi}} \right] d\eta =$$

$$= \frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \sqrt{h_{\text{ext}}} \int_0^{\infty} e^{-\eta^2} d\eta - \frac{\eta}{\sqrt{\pi}} \int_0^{\infty} e^{-\eta^2} d\eta =$$

$$= \frac{\psi_0}{k} \frac{2}{\sqrt{\pi}} \sqrt{h_{\text{ext}}} \left(-\frac{1}{2} e^{-\frac{\eta^2}{\sqrt{\pi}}} \right) - \frac{\eta}{\sqrt{\pi}} \int_0^{\infty} e^{-\eta^2} d\eta =$$

$$I = \int_{-\infty}^{\infty} \left(\frac{\sqrt{4\alpha t}}{\pi} \exp(-y^2/4\alpha t) - \frac{2y}{\pi} \int_0^{\infty} e^{-\eta^2} d\eta \right) dy$$

$$\frac{\partial I}{\partial t} = \alpha \frac{\partial^2 I}{\partial y^2}$$

$T = (t, y)$ disregarding $\frac{\partial I}{\partial y}$

$$\frac{\partial I}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\sqrt{4\alpha t}}{\pi} e^{-y^2/4\alpha t} \right)$$

$$\frac{\partial I}{\partial t} = \frac{\partial I}{\partial u} \frac{\partial u}{\partial t} \rightarrow \frac{\partial}{\partial t} \left(\frac{\sqrt{4\alpha t}}{\pi} \right) = \frac{4\alpha}{\pi} \cdot \frac{1}{2\sqrt{4\alpha t}} = \frac{4\alpha}{\pi} \cdot \frac{1}{2\sqrt{4\alpha t}}$$

$$\frac{\partial I}{\partial t} = \frac{4\pi}{\pi} \frac{\partial I}{\partial u} \rightarrow \frac{\partial \sqrt{u}}{\partial u} = \frac{1}{2\sqrt{u}}$$

$$\frac{\partial I}{\partial t} = \frac{4\alpha}{\pi} \cdot \frac{1}{2\sqrt{4\alpha t}} = \frac{4\alpha}{\pi} \cdot \frac{1}{2\sqrt{4\alpha t}}$$

$$= \frac{4\alpha}{\pi} \cdot \frac{1}{2\sqrt{4\alpha t}} = \frac{4\alpha}{\pi} \cdot \frac{1}{2\sqrt{4\alpha t}}$$

$$= \frac{\alpha^{1/2}}{\pi^{1/2} \sqrt{4t}} = \sqrt{\frac{\alpha}{\pi t}}$$

$$\frac{\partial I}{\partial t} = \frac{\partial I}{\partial u} \frac{\partial u}{\partial t} \rightarrow \frac{\partial}{\partial t} \left(\frac{\sqrt{4\alpha t}}{\pi} e^{-y^2/4\alpha t} \right) = \frac{\partial}{\partial t} \left(\frac{\sqrt{4\alpha t}}{\pi} \right) \cdot e^{-y^2/4\alpha t} + \frac{\sqrt{4\alpha t}}{\pi} \cdot \frac{\partial}{\partial t} \left(e^{-y^2/4\alpha t} \right)$$

Product rule

$$\frac{\partial I}{\partial t} = \left(\sqrt{\frac{\alpha}{\pi t}} e^{-y^2/4\alpha t} - \sqrt{\frac{4\alpha t}{\pi}} \frac{y^2}{4\alpha t^2} e^{-y^2/4\alpha t} \right) = \sqrt{\frac{\alpha}{\pi t}} e^{-y^2/4\alpha t} - \frac{y^2}{4\alpha t^2} \sqrt{\frac{4\alpha t}{\pi}} e^{-y^2/4\alpha t}$$

$$I = \int_{-\infty}^{\infty} \left(\frac{\sqrt{4\alpha t}}{\pi} e^{-y^2/4\alpha t} - \frac{y^2}{4\alpha t^2} \sqrt{\frac{4\alpha t}{\pi}} e^{-y^2/4\alpha t} \right) dy = \frac{1}{\sqrt{4\alpha t}} \int_{-\infty}^{\infty} e^{-y^2/4\alpha t} dy - \frac{y^2}{4\alpha t^2} \int_{-\infty}^{\infty} \sqrt{\frac{4\alpha t}{\pi}} e^{-y^2/4\alpha t} dy$$

$$\frac{\partial^2 T}{\partial y^2} = \sqrt{\frac{h_{\text{ext}}}{\pi}} \left(-\frac{y}{2\alpha t} e^{-\frac{y^2}{4\alpha t}} \right) =$$

For average
part of eq.

$$\frac{\partial T}{\partial t} = \left(\frac{\alpha}{\sqrt{\pi t}} + \frac{y^2}{4\pi\alpha t^2} \right) e^{-\frac{y^2}{4\alpha t}}$$

$$\alpha \frac{\partial^2 T}{\partial y^2} = \left(\frac{y^2}{4\pi\alpha t^2} - \frac{2\alpha t}{4\pi\alpha t^2} \right) e^{-\frac{y^2}{4\alpha t}} = \left(\frac{y^2}{4\pi\alpha t^2} - \left(\frac{\alpha}{\pi t} \right) \right) e^{-\frac{y^2}{4\alpha t}}$$

Then, $\frac{\partial T}{\partial t} = \left(\left(\frac{1}{4\pi\alpha t} \right) \frac{y^2}{t} \right) e^{-\frac{y^2}{4\alpha t}}$

$$\alpha \frac{\partial^2 T}{\partial y^2} = \left(-2\sqrt{\frac{\alpha}{\pi t}} + \sqrt{\frac{1}{4\pi\alpha t}} \frac{y^2}{t} \right) e^{-\frac{y^2}{4\alpha t}}$$

$$\frac{\partial T}{\partial t} + \frac{\partial^2 T}{\partial y^2} = \alpha \frac{\partial^2 T}{\partial y^2} + \alpha \frac{\partial^2 T}{\partial y^2}$$

b.c. (i) $y = \infty$ $T = T_0$ and $t \rightarrow \infty$

$$T - T_0 = \frac{T_0}{L} \left(\frac{h_{\text{ext}}}{\pi} \exp^{-\infty} \right) - 2 \operatorname{erf} \left(\frac{\infty}{h_{\text{ext}}} \right)$$

$$T = T_0$$

b.c. as $y \rightarrow \infty$ $q = 0$

$$q_y = q_0 \operatorname{erfc} \left(\frac{\infty}{h_{\text{ext}}} \right)$$

$$q_y = 0$$